

GCSE Mathematics - Foundation Tier

Paper 2 (Calculator) Practice - Set 1

Time: 1 hour 30 minutes | Total marks: 80 | Calculator allowed

Instructions

- Use black ink or ball-point pen.
- Answer ALL questions in the spaces provided.
- You must show all your working - answers given without working may not score full marks.
- Diagrams are NOT accurately drawn unless otherwise stated.
- If your calculator does not have a pi button, use $\pi = 3.142$.

1.

Write these numbers in order of size. Start with the smallest.

0.7

0.07

0.605

0.65

(Total for Question 1 is 1 marks)

2.

Work out 15% of £240.

(Total for Question 2 is 2 marks)

3.

- (a) Convert 3.5 km to metres. (1)
- (b) Convert 4500 g to kilograms. (1)

(Total for Question 3 is 2 marks)

4.

Write the ratio 24 : 36 in its simplest form.

(Total for Question 4 is 2 marks)

5.

A class of 50 students were asked their favourite drink. The bar chart shows the results.

[Diagram: bar chart of favourite drinks for 50 students. Bar heights - Tea: 10, Coffee: 15, Water: 12, Juice: 13.]

- (a) Write down the number of students who chose Coffee. (1)
- (b) Work out the percentage of students who chose Water. (2)

(Total for Question 5 is 3 marks)

6.

Simplify $5a + 3b - 2a + 7b$.

(Total for Question 6 is 2 marks)

7.

A car travels 84 km in 1 hour 12 minutes.

Work out the average speed of the car in kilometres per hour.

(Total for Question 7 is 3 marks)

8.

The table shows the number of pets owned by 24 students.

Number of pets	Frequency
0	5
1	8
2	6
3	3
4	2

Work out the mean number of pets per student. Give your answer to 2 decimal places.

(Total for Question 8 is 3 marks)

9.

Anil, Beth and Carla share £450 in the ratio 2 : 3 : 4.

Work out how much each person receives.

(Total for Question 9 is 3 marks)

10.

Solve $5x - 7 = 3x + 11$.

You must show all your working.

(Total for Question 10 is 3 marks)

11.

[Diagram: right-angled triangle with shorter sides 6 cm and 8 cm; hypotenuse labelled x cm.]

Work out the length x . Give your answer in centimetres.

(Total for Question 11 is 3 marks)

12.

The pie chart shows the favourite sport of 90 people.

[Diagram: pie chart with sectors - Football: 160 deg, Cricket: 80 deg, Tennis: 60 deg, Rugby: 60 deg.]

- (a) Work out the number of people who chose Football. (2)
- (b) What fraction of people chose Tennis? Give your answer in its simplest form. (2)

(Total for Question 12 is 4 marks)

13.

A formula for converting temperatures from degrees Celsius (C) to degrees Fahrenheit (F) is

$$F = 1.8C + 32$$

- (a) Work out F when C = 25. (2)

The area A of a triangle with base b and perpendicular height h is given by

$$A = 0.5 \times b \times h$$

- (b) Work out A when b = 14 cm and h = 9 cm. (2)

(Total for Question 13 is 4 marks)

14.

In 2024, the price of a laptop was £680. In 2026, the price of the same laptop has increased to £765.

- (a) Work out the percentage increase in price. Give your answer to 1 decimal place. (3)

A different laptop has a sale price of £540 after a 10% discount has been applied to the original price.

- (b) Work out the original price of this laptop. (2)

(Total for Question 14 is 5 marks)

15.

Hannah invests £4000 in a savings account. The account pays compound interest at a rate of 2.5% per year.

Work out the value of her investment after 3 years. Give your answer to the nearest penny.

(Total for Question 15 is 4 marks)

16.

A recipe for 12 biscuits uses 180 g flour, 90 g butter and 75 g sugar.

- (a) Work out the amount of flour needed for 20 biscuits. (2)

Saskia has 600 g of flour, 300 g of butter and 250 g of sugar.

- (b) Work out the largest number of complete biscuits Saskia can make. You must show your working. (3)

(Total for Question 16 is 5 marks)

17.

The table shows the number of hours of revision done by 8 students before a maths test, and their test scores out of 100.

Hours	1	2	3	4	5	6	7	8
Score	35	42	50	58	62	70	78	84

- (a) On the grid provided, plot these points to form a scatter graph. (2)
- (b) Describe the correlation shown by your scatter graph. (1)
- (c) Draw a line of best fit on your scatter graph. (1)
- (d) A student studied for 4.5 hours. Use your line of best fit to estimate the score. (1)

(Total for Question 17 is 5 marks)

18.

A prism has a triangular cross-section. The triangle has a base of 8 cm and a perpendicular height of 5 cm. The length of the prism is 15 cm.

[Diagram: triangular prism with cross-section base 8 cm, height 5 cm; length of prism 15 cm.]

- (a) Work out the volume of the prism. (3)

The prism is made of a metal with density 7.8 g per cubic centimetre.

- (b) Work out the mass of the prism. Give your answer in kilograms. (2)

(Total for Question 18 is 5 marks)

19.

In a school survey, 72 students were asked to choose their favourite subject. The results are shown below.

Subject	Frequency
Maths	24
English	12
Science	18
Art	9
Other	9

Draw an accurate pie chart to represent this information. Label each sector clearly.

(Total for Question 19 is 5 marks)

20.

A rectangle has a length of $(3x + 4)$ cm and a width of $(x + 2)$ cm.

[Diagram: rectangle of length $(3x + 4)$ cm and width $(x + 2)$ cm.]

The perimeter of the rectangle is 60 cm.

- (a) Show that $8x + 12 = 60$. (2)
- (b) Solve your equation to find the value of x . (2)
- (c) Hence write down the length and the width of the rectangle. (1)

(Total for Question 20 is 5 marks)

21.

Marcus is travelling from the UK to Spain. The exchange rate is £1 = €1.18.

- (a) Convert £450 into euros. (2)

While in Spain, Marcus sees a watch on sale for €295. The identical watch is on sale in the UK for £260.

- (b) In which country is the watch cheaper? You must show all your working. (4)

(Total for Question 21 is 6 marks)

22.

A spinner is divided into 5 equal sections, coloured red, blue, green, yellow and white. The spinner is fair.

- (a) Write down the probability that, in a single spin, the spinner lands on green. (1)

The spinner is now spun twice. The two spins are independent.

- (b) List all the possible outcomes when the spinner lands on red on the first spin. (1)

- (c) Work out the probability that the spinner lands on red on the first spin and yellow on the second spin. (3)

(Total for Question 22 is 5 marks)

TOTAL FOR PAPER: 80 MARKS

Mark Scheme

Foundation Tier - Paper 2 (Calculator) - Set 1 | 80 marks

Question 1

- Correct order: 0.07, 0.605, 0.65, 0.7 - B1

Total: 1 marks

Question 2

- 240×0.15 or $240 \times 15 / 100$ - M1
- £36 - A1

Total: 2 marks

Question 3

- (a) 3500 m - B1
- (b) 4.5 kg - B1

Total: 2 marks

Question 4

- Both parts divided by a common factor (e.g. 12 : 18) - M1
- 2 : 3 - A1

Total: 2 marks

Question 5

- (a) 15 - B1
- (b) $12 / 50$ or $12 / 50 \times 100$ - M1
- (b) 24 (%) - A1

Total: 3 marks

Question 6

- At least one correct collected term (3a or 10b) - M1
- $3a + 10b$ - A1

Total: 2 marks

Question 7

- 1 hr 12 min written as 1.2 hr or 72 min - M1
- $84 / 1.2$ or $84 / 72 \times 60$ - M1
- 70 (km/h) - A1

Total: 3 marks

Question 8

- Sum(fx) attempted with at least 3 correct products from {0, 8, 12, 9, 8} - M1
- Sum(fx) / Sum(f) used ($37 / 24$) - M1
- 1.54 - A1

Total: 3 marks

Question 9

- $450 / 9 = 50$ - M1
- Method of multiplying by 2, 3 and 4 - M1
- Anil £100, Beth £150, Carla £200 - A1

Total: 3 marks

Question 10

- Variable terms collected on one side ($2x = 18$) - M1
- Numerical terms collected on the other side - M1
- $x = 9$ - A1

Total: 3 marks

Question 11

- Pythagoras applied: $6^2 + 8^2$ - M1
- $36 + 64 = 100$ - M1
- $x = 10$ (cm) - A1

Total: 3 marks

Question 12

- (a) $160 / 360 \times 90$ - M1
- (a) 40 - A1
- (b) $60 / 360$ seen - M1
- (b) $1/6$ - A1

Total: 4 marks

Question 13

- (a) $1.8 \times 25 = 45$ - M1
- (a) $F = 77$ - A1
- (b) $0.5 \times 14 \times 9$ - M1
- (b) $A = 63$ - A1

Total: 4 marks

Question 14

- (a) $765 - 680 = 85$ - M1
- (a) $85 / 680 \times 100$ - M1
- (a) 12.5 (%) - A1
- (b) $540 / 0.9$ or $540 / 90 \times 100$ - M1
- (b) £600 - A1

Total: 5 marks

Question 15

- 1.025 seen - M1
- $4000 \times 1.025^2 (= 4202.50)$ seen - M1
- 4000×1.025^3 - M1
- £4307.56 - A1

Total: 4 marks

Question 16

- (a) $180 / 12 \times 20$ - M1
- (a) 300 (g) - A1
- (b) Per-biscuit amounts (15, 7.5, 6.25) or $600/15 = 40$ seen - M1
- (b) All three ingredients checked - M1
- (b) 40 biscuits - A1

Total: 5 marks

Question 17

- (a) 8 points plotted correctly - B2 (B1 for at least 6 correct)
- (b) Positive (correlation) - B1
- (c) Sensible line of best fit drawn - B1
- (d) Score read at 4.5 hours (accept 53 - 62) - B1

Total: 5 marks

Question 18

- (a) $0.5 \times 8 \times 5 = 20$ (area of triangle) - M1
- (a) 20×15 - M1
- (a) 300 cm^3 - A1
- (b) $7.8 \times 300 (= 2340 \text{ g})$ - M1
- (b) 2.34 (kg) - A1

Total: 5 marks

Question 19

- $360 / 72 = 5$ (angle per student) - M1
- At least 3 angles correct: Maths 120, English 60, Science 90, Art 45, Other 45 (deg) - M1
- All 5 angles correct - A1
- Pie chart drawn within 2 deg tolerance - B1
- All sectors labelled - B1

Total: 5 marks

Question 20

- (a) $2(3x + 4) + 2(x + 2)$ or $6x + 8 + 2x + 4$ - M1
- (a) $8x + 12 = 60$ derived - C1
- (b) $8x = 48$ - M1
- (b) $x = 6$ - A1
- (c) Length 22 cm and width 8 cm - B1

Total: 5 marks

Question 21

- (a) 450×1.18 - M1
- (a) €531 - A1
- (b) Method: $295 / 1.18$ or 260×1.18 - M1
- (b) Conversion correctly completed (£250 or €306.80) - M1
- (b) Comparison stated between converted values - M1
- (b) Spain is cheaper, supported by working - A1

Total: 6 marks

Question 22

- (a) $\frac{1}{5}$ - B1
- (b) RR, RB, RG, RY, RW - B1
- (c) Multiplication chosen / events independent - M1
- (c) $\frac{1}{5} \times \frac{1}{5}$ - M1
- (c) $\frac{1}{25}$ - A1

Total: 5 marks

Worked Solutions

Foundation Tier - Paper 2 (Calculator) - Set 1 | Detailed explanations

Question 1

To compare decimals, line up the decimal points and look at digits from left to right.

Pad each number to three decimal places: 0.700, 0.070, 0.605, 0.650.

Compare tenths first: 0.0 (smallest), then two 0.6s, then 0.7 (largest).

For the two 0.6s look at hundredths: 0.605 (0) < 0.650 (5).

Final order: 0.07, 0.605, 0.65, 0.7.

Question 2

"15% of £240" means 15 hundredths of 240.

Convert the percentage: $15 / 100 = 0.15$.

Multiply: $0.15 \times 240 = 36$.

Answer: £36.

Mental check - 10% = £24, 5% = £12, total £36.

Question 3

(a) 1 km = 1000 m, so to go from km to m multiply by 1000.

$3.5 \times 1000 = 3500$ m.

(b) 1000 g = 1 kg, so to go from g to kg divide by 1000.

$4500 / 1000 = 4.5$ kg.

Question 4

To simplify a ratio, divide both numbers by their highest common factor (HCF).

Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24.

Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36.

HCF = 12. Divide both parts: $24/12 = 2$ and $36/12 = 3$.

Answer: 2 : 3.

Question 5

(a) Read the Coffee bar directly - 15 students.

(b) Water = 12 out of 50.

Percentage = $(12 / 50) \times 100 = 0.24 \times 100 = 24\%$.

Answer: 24%.

Question 6

Collect like terms (same letter).

a-terms: $5a - 2a = 3a$.

b-terms: $3b + 7b = 10b$.

Answer: $3a + 10b$.

Question 7

Speed = distance / time.

Distance = 84 km.

Time = 1 h 12 min. Convert minutes: $12 / 60 = 0.2$ h.

So time = 1.2 h.

Speed = $84 / 1.2 = 70$.

Answer: 70 km/h.

Question 8

Mean from a frequency table = $\text{Sum}(fx) / \text{Sum}(f)$.

fx for each row: $0 \times 5 = 0$, $1 \times 8 = 8$, $2 \times 6 = 12$, $3 \times 3 = 9$, $4 \times 2 = 8$.

$\text{Sum}(fx) = 0 + 8 + 12 + 9 + 8 = 37$.

$\text{Sum}(f) = 5 + 8 + 6 + 3 + 2 = 24$.

Mean = $37 / 24 = 1.5416\ldots$

Rounded to 2 dp: 1.54.

Question 9

Total parts = $2 + 3 + 4 = 9$.

One part = $\pounds 450 / 9 = \pounds 50$.

Anil = $2 \times \pounds 50 = \pounds 100$.

Beth = $3 \times \pounds 50 = \pounds 150$.

Carla = $4 \times \pounds 50 = \pounds 200$.

Check: $100 + 150 + 200 = \pounds 450$.

Question 10

Collect x-terms on one side, numbers on the other.

$5x - 7 = 3x + 11$.

Subtract $3x$: $2x - 7 = 11$.

Add 7: $2x = 18$.

Divide by 2: $x = 9$.

Check: $5(9) - 7 = 38$ and $3(9) + 11 = 38$.

Question 11

Pythagoras: $\text{hypotenuse}^2 = \text{side}_1^2 + \text{side}_2^2$.

$x^2 = 6^2 + 8^2 = 36 + 64 = 100$.

$x = \sqrt{100} = 10$.

Answer: $x = 10$ cm.

Question 12

(a) Football = 160 deg out of 360 deg.

$$\text{Number} = (160 / 360) \times 90 = 14400 / 360 = 40.$$

Answer: 40 people.

(b) Tennis = 60 deg out of 360 deg.

$$\text{Fraction} = 60/360 = 1/6.$$

Answer: 1/6.

Question 13

(a) $F = 1.8C + 32$, with $C = 25$.

$$F = 1.8 \times 25 + 32 = 45 + 32 = 77.$$

(b) $A = 0.5 \times b \times h$, with $b = 14$, $h = 9$.

$$A = 0.5 \times 14 \times 9 = 7 \times 9 = 63.$$

Answer: $A = 63 \text{ cm}^2$.

Question 14

(a) Percentage change = (change / original) $\times$ 100.

$$\text{Change} = 765 - 680 = \text{£}85. \text{ Original} = \text{£}680.$$

$$\% \text{ increase} = (85 / 680) \times 100 = 12.5\%.$$

(b) After 10% off, sale price = 90% of original.

$$0.9 \times \text{original} = 540, \text{ so original} = 540 / 0.9 = \text{£}600.$$

Question 15

$$\text{Multiplier} = 1 + 2.5/100 = 1.025.$$

$$\text{Year 1: } 4000 \times 1.025 = \text{£}4100.00.$$

$$\text{Year 2: } 4100 \times 1.025 = \text{£}4202.50.$$

$$\text{Year 3: } 4202.50 \times 1.025 = \text{£}4307.5625.$$

$$\text{Or directly: } 4000 \times 1.025^3 = 4307.5625.$$

Rounded to the nearest penny: £4307.56.

Question 16

(a) Per biscuit, flour = $180 / 12 = 15 \text{ g}$.

$$\text{For 20 biscuits: } 15 \times 20 = 300 \text{ g}.$$

(b) Per biscuit: flour 15 g, butter 7.5 g, sugar 6.25 g.

$$\text{Flour limit: } 600 / 15 = 40 \text{ biscuits}.$$

$$\text{Butter limit: } 300 / 7.5 = 40 \text{ biscuits}.$$

$$\text{Sugar limit: } 250 / 6.25 = 40 \text{ biscuits}.$$

Smallest of these is the maximum number of biscuits: 40.

Question 17

- (a) Plot 8 points (1,35), (2,42), (3,50), (4,58), (5,62), (6,70), (7,78), (8,84).
 - (b) As hours increase, score increases $\rightarrow$ positive correlation.
 - (c) Draw a straight line through the middle of the cloud of points.
 - (d) Read off the score at $x = 4.5$: around 57 - 60 (depending on line drawn).
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Question 18

- (a) Area of triangle = $0.5 \times 8 \times 5 = 20 \text{ cm}^2$.
Volume of prism = area $\times$ length = $20 \times 15 = 300 \text{ cm}^3$.
 - (b) Mass = density $\times$ volume = $7.8 \times 300 = 2340 \text{ g}$.
Convert: $2340 / 1000 = 2.34 \text{ kg}$.
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Question 19

- Angle per student = $360 / 72 = 5 \text{ deg}$.
Maths: $24 \times 5 = 120 \text{ deg}$.
English: $12 \times 5 = 60 \text{ deg}$.
Science: $18 \times 5 = 90 \text{ deg}$.
Art: $9 \times 5 = 45 \text{ deg}$.
Other: $9 \times 5 = 45 \text{ deg}$.
Total check: $120 + 60 + 90 + 45 + 45 = 360 \text{ deg}$.
Use a protractor to draw each sector and label every one.
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Question 20

- (a) Perimeter = $2(3x + 4) + 2(x + 2) = 6x + 8 + 2x + 4 = 8x + 12$.
Setting equal to 60 gives $8x + 12 = 60$. (Shown.)
 - (b) $8x + 12 = 60 \rightarrow 8x = 48 \rightarrow x = 6$.
 - (c) Length = $3(6) + 4 = 22 \text{ cm}$. Width = $6 + 2 = 8 \text{ cm}$.
Check the perimeter: $2(22) + 2(8) = 60 \text{ cm}$.
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Question 21

- (a) $\text{£}450 \times 1.18 = \text{€}531$.
 - (b) Method 1 - put both prices in pounds: $\text{€}295 / 1.18 = \text{£}250$.
UK $\text{£}260$ vs Spain $\text{£}250 \rightarrow$ Spain is $\text{£}10$ cheaper.
Method 2 - put both prices in euros: $\text{£}260 \times 1.18 = \text{€}306.80$.
Spain $\text{€}295$ vs UK $\text{€}306.80 \rightarrow$ Spain is $\text{€}11.80$ cheaper.
Conclusion: the watch is cheaper in Spain.
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Question 22

- (a) 5 equal sections, 1 green: $P(\text{green}) = 1/5$.
- (b) RR, RB, RG, RY, RW.

(c) Spins are independent. Multiply individual probabilities.

$P(\text{red, then yellow}) = \frac{1}{5} \times \frac{1}{5} = \frac{1}{25}$.
